

Budget Justification

Senior/Key Personnel (Line A)

Salaries for two senior/key personnel are requested. All proposing personnel are new employees of the FreeMoCap Foundation; requested salary rates were set by the Foundation's Board in its adopted written compensation policy and are consistent with that policy. NSF does not recognize executive titles for salary-rate determination; the rates below therefore correspond to each individual's functional responsibilities, and each is well below the 75th-percentile Bureau of Labor Statistics rate for the corresponding occupation in the geographic area where the work will be carried out (standard occupational classification codes and BLS references listed per person).

Jonathan Matthis — Principal Investigator

- **Functional title / SOC:** Computer and Information Research Scientist (SOC 15-1221); BLS OEWS reference: bls.gov/oes/current/oes151221.htm
- **Salary rate:** \$75,000/year; **Effort:** 12.0 calendar months; **Requested:** \$75,000
- **Responsibilities:** Overall scientific and organizational direction of the project. Leads governance scoping (comparison and evaluation of stewardship models), the risk-analysis and security plan, and the core-versus-extension architecture decisions. Serves as Technical Lead for the mandatory I-Corps for PESOSE training, with decision-making authority over project direction.

Aaron Cherian — co-Principal Investigator

- **Functional title / SOC:** Software Developer (SOC 15-1252); BLS OEWS reference: bls.gov/oes/current/oes151252.htm
- **Salary rate:** \$75,000/year; **Effort:** 9.0 calendar months; **Requested:** \$56,250
- **Responsibilities:** Leads the technical scoping work: definition of the community-extension architecture, design of the validation, benchmarking, and regression-testing infrastructure, and coordination of the data-quality and provenance metadata work. Serves as Entrepreneurial Lead for the I-Corps for PESOSE training, coordinating stakeholder outreach, discovery interviews, and weekly reporting. The remaining 3 months of his effort continue on unfunded Foundation activities.

Other Personnel (Line B)

Paul Matthis — Operations Lead

- **Functional title / SOC:** Network and Computer Systems Administrator (SOC 15-1244); BLS OEWS reference: bls.gov/oes/current/oes151244.htm
- **Salary rate:** \$50,000/year; **Effort:** 7.0 calendar months; **Requested:** \$29,167
- **Responsibilities:** Maintains the technical infrastructure on which the planning activities run: release and continuous-integration infrastructure (GitHub Actions), project web services, and the Google Cloud Platform environment hosting project telemetry, including the provenance and data-quality instrumentation scoped in the Risk Analysis and Security Plan. Provides logistics support for the community education pilots.

Fringe Benefits (Line C)

Fringe benefits are requested at a rate of 20% of salaries, consisting of statutory payroll taxes (7.65%), employer contributions toward individual health insurance coverage, and a modest retirement plan contribution. Fringe rates will be established in conjunction with the Foundation's move

to W-2 employment (see Professional Services under Other Direct Costs) and applied uniformly to all project personnel.

Equipment (Line D)

No equipment is requested.

Travel (Line E)

Travel is requested to sustain the community-building and ecosystem-discovery activities central to Track 1: \$7,000 total, covering two domestic scientific conferences (estimated \$3,500 per trip: registration, airfare, four nights lodging, per diem, ground transportation):

- **American Society of Biomechanics (ASB) annual meeting** — engagement with the biomechanics research community that constitutes FreeMoCap’s core academic user base; dissemination of project results and recruitment of contributors and ecosystem partners.
- **Society for Neuroscience (SfN) annual meeting** — engagement with the neuroscience research community, a major user population for accessible motion measurement, and a venue for ecosystem discovery interviews with prospective users and contributors.

No international travel is requested.

Participant Support (Line F)

\$5,000 is requested to provide participant stipends for community members in the education and onboarding pilots conducted with partner organizations, lowering barriers to participation for learners and prospective contributors from under-resourced communities. Participant support costs are excluded from the indirect-cost base.

Other Direct Costs (Line G)

Materials and Supplies — \$8,000

Multi-camera capture kits (consumer-grade webcams and mounting hardware) and workshop materials for the hands-on education pilots with partner organizations, and for hardware-configuration testing across supported platforms.

I-Corps for PESOSE — \$10,000

Participation in the mandatory I-Corps for PESOSE training is budgeted at \$10,000, within the \$30,000 program ceiling:

- I-Corps participation fee: \$2,500
- Industry Mentor stipend: \$3,000 (Endurance Dahan, experienced open-source ecosystem leader; supports the team’s discovery work)
- Ecosystem-discovery travel within the training window: \$4,500

The training team comprises the Technical Lead (Matthis), Entrepreneurial Lead (Cherian), and Industry Mentor. No gifts, meals, marketing expenses, or survey tools are requested.

Professional Services (Contractual) — \$7,500

Human resources and payroll administration services required to employ the project personnel as W-2 employees and administer the benefits described under Fringe Benefits. This is a direct requirement of carrying out the proposed effort: the personnel effort budgeted above cannot be executed without employer-of-record, payroll, and benefits administration infrastructure.

No consultant services and no subawards are requested. The Foundation is a nonprofit organization; no owners or equity holders exist whose compensation would be restricted.

Indirect Costs (Line I)

The FreeMoCap Foundation does not have a federally negotiated indirect cost rate agreement and elects the 15% de minimis rate on modified total direct costs. MTDC = \$255,000 (total direct costs of \$260,000 excluding \$5,000 participant support costs). Indirect costs: \$38,250.

Budget Summary

Line	Amount
A. Senior/Key Personnel	\$185,417
B. Other Personnel	\$29,167
C. Fringe Benefits (20%)	\$37,083
D. Equipment	\$0
E. Travel	\$7,000
F. Participant Support	\$5,000
G. Other Direct Costs	\$25,500
H. Total Direct Costs	\$260,000
I. Indirect Costs (15% MTDC)	\$38,250
J. Total Requested	\$298,250